

**Angelo Cantoli - Simone Edmondo  
Bedini**



**POLITECNICO**  
MILANO 1863

# **Design Document**

---

|                       |                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Deliverable:</b>   | RASD                                                                                                          |
| <b>Title:</b>         | Design Document                                                                                               |
| <b>Authors:</b>       | Angelo Cantoli, Simone Edmondo Bedini                                                                         |
| <b>Version:</b>       | 1.0                                                                                                           |
| <b>Date:</b>          | 28/10/2025                                                                                                    |
| <b>Download page:</b> | <a href="https://github.com/angelo-cantoli/CantoliBedini">https://github.com/angelo-cantoli/CantoliBedini</a> |
| <b>Copyright:</b>     | Copyright © 2025 - Angelo Cantoli, Simone Edmondo Bedini - All rights reserved                                |

---

## Contents

|                                                    |           |
|----------------------------------------------------|-----------|
| <b>Table of Contents</b>                           | <b>3</b>  |
| <b>List of Figures</b>                             | <b>4</b>  |
| <b>List of Tables</b>                              | <b>4</b>  |
| <b>1 Introduction</b>                              | <b>5</b>  |
| 1.1 Purpose                                        | 5         |
| 1.2 Scope                                          | 5         |
| 1.3 Definitions, Acronyms, Abbreviations           | 5         |
| 1.4 Reference Documents                            | 5         |
| 1.5 Document Structure                             | 5         |
| <b>2 Architectural Design</b>                      | <b>6</b>  |
| <b>3 User Interface Design</b>                     | <b>7</b>  |
| <b>4 Requirements Traceability</b>                 | <b>8</b>  |
| <b>5 Implementation, Integration and Test Plan</b> | <b>9</b>  |
| <b>6 Effort Spent</b>                              | <b>10</b> |
| <b>7 References</b>                                | <b>11</b> |

List of Figures

List of Tables

1     Definitions, Acronyms and Abbreviations     . . . . . 5

# 1 Introduction

## 1.1 Purpose

BBP is a software designed to simplify the life of everyone who goes on bike trips and wants to enjoy them without worrying about finding the best routes.

It is intended as a community for cyclists, in which every person can share information about their personal routes, so that everyone may take advantage of the information and experience of other bikers. The application is intended for smartphone, where it can make use of GPS and sensor data to seamlessly reconstruct the path a cyclist has taken.

## 1.2 Scope

## 1.3 Definitions, Acronyms, Abbreviations

| Acronym | Definition                  |
|---------|-----------------------------|
| BBP     | Best Bike Paths application |
| GPS     | Global Positioning System   |
| UI      | User Interface              |
| SP      | Shared Phenomenon           |
| WP      | World Phenomenon            |
| UC      | Use Case                    |
| DA      | Domain Assumption           |
| G       | Goal                        |

Table 1: Definitions, Acronyms and Abbreviations

## 1.4 Reference Documents

## 1.5 Document Structure

## **2 Architectural Design**

### **3 User Interface Design**

## **4 Requirements Traceability**

## **5 Implementation, Integration and Test Plan**

## **6 Effort Spent**

## **7 References**